

Drone-based AI-driven Parking Pass Detection And Enforcement System

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Abstract

- Parking enforcement is still done on foot as officer walks and read every windshield pass and write tickets by hand.
- This system automates the whole process, a drone captures the ariel footage of the parked cars, AI detects and reads the parking pass, instantly flagging the pass as valid or expired.
- Violations trigger an evidence photo and other details which are stored in cloud to be reviewed by admin.

Objectives

- To create a Privacy-preserving, non-invasive smart system with the potential to adapt advanced technologies.
- To do all the processing while still staying as efficient and accurate as possible.

Challenges

- The signal does not reach outside the building and is cut off.
- Sunlight reflecting off the windows can make outdoor pass detection difficult at certain times of day.

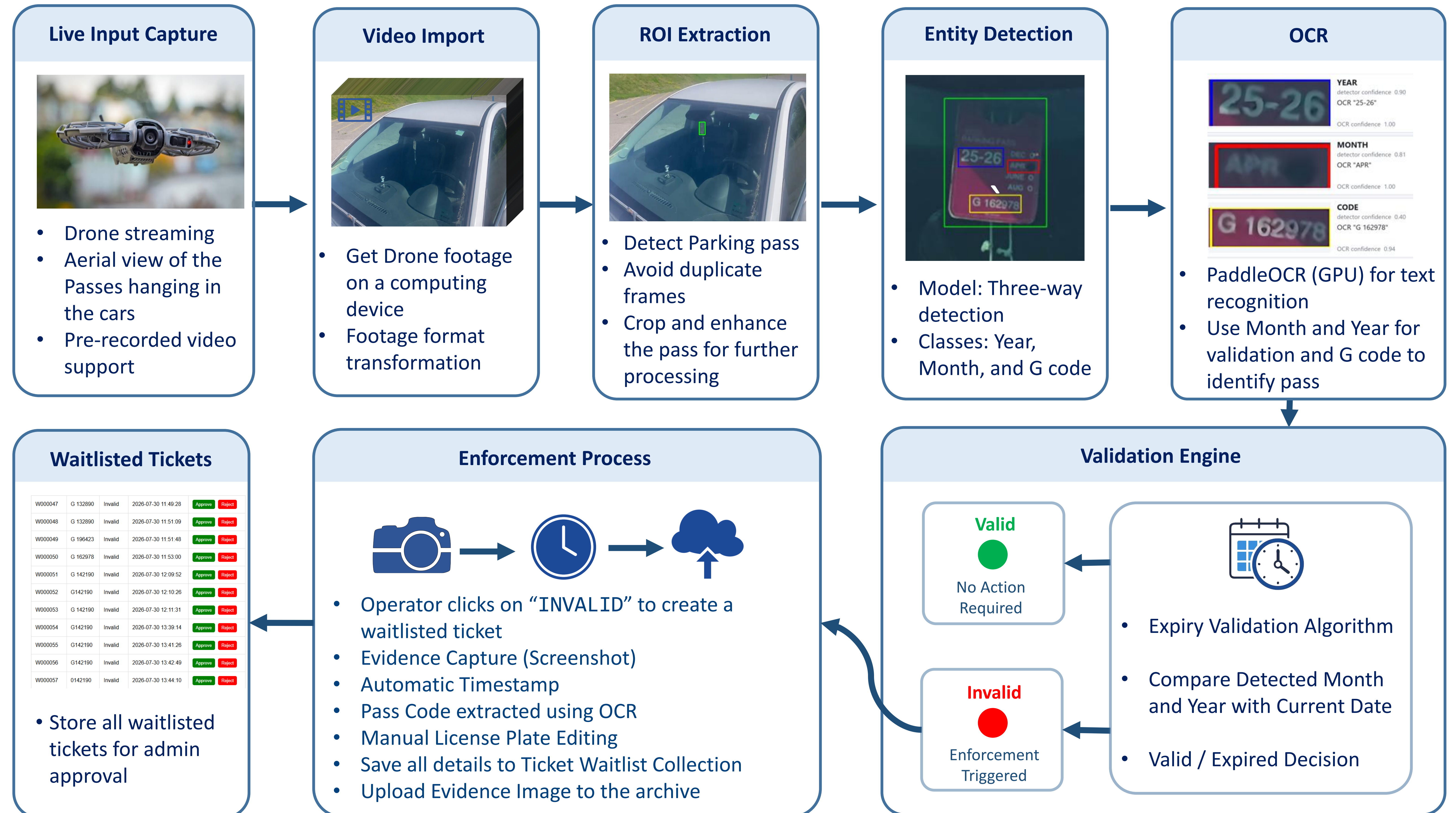
Current Results

Samples	ROI Detection	Decision accuracy (OCR based)
45	164/180 (91.1%)	37/45 (82.2%)

Conclusion

- The system is feasible and shows the potential for real-world use. With further improvements, it could become a reliable parking pass verification system.

The Methodology of the Proposed Drone-based AI-driven Solution



The Enforcement Dashboard: Waitlist Review, Ticket Generation, and Automated Emailing

